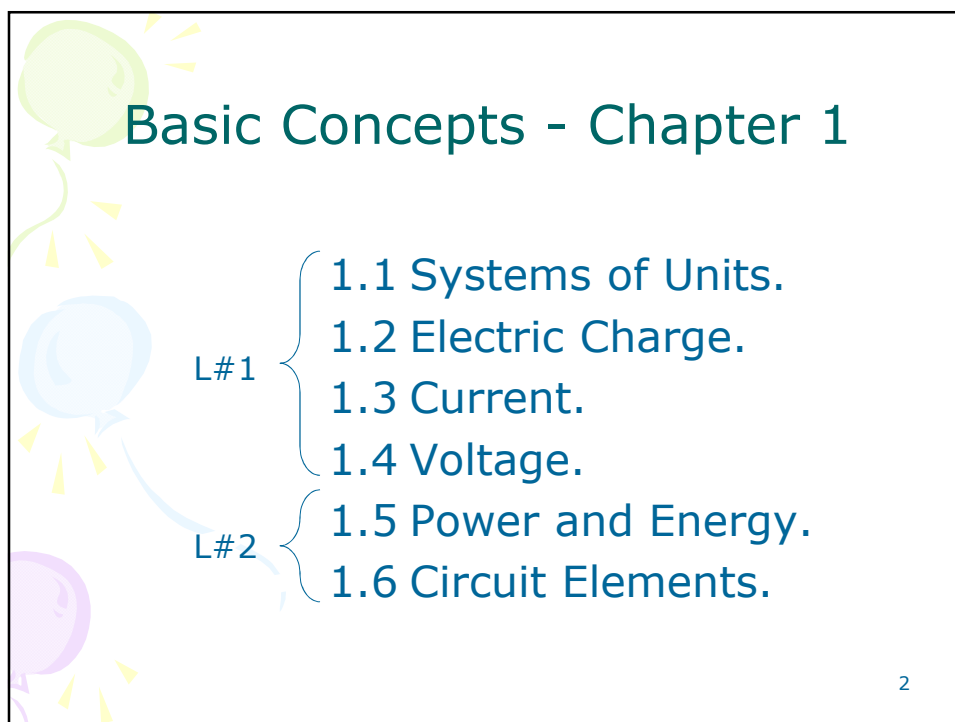
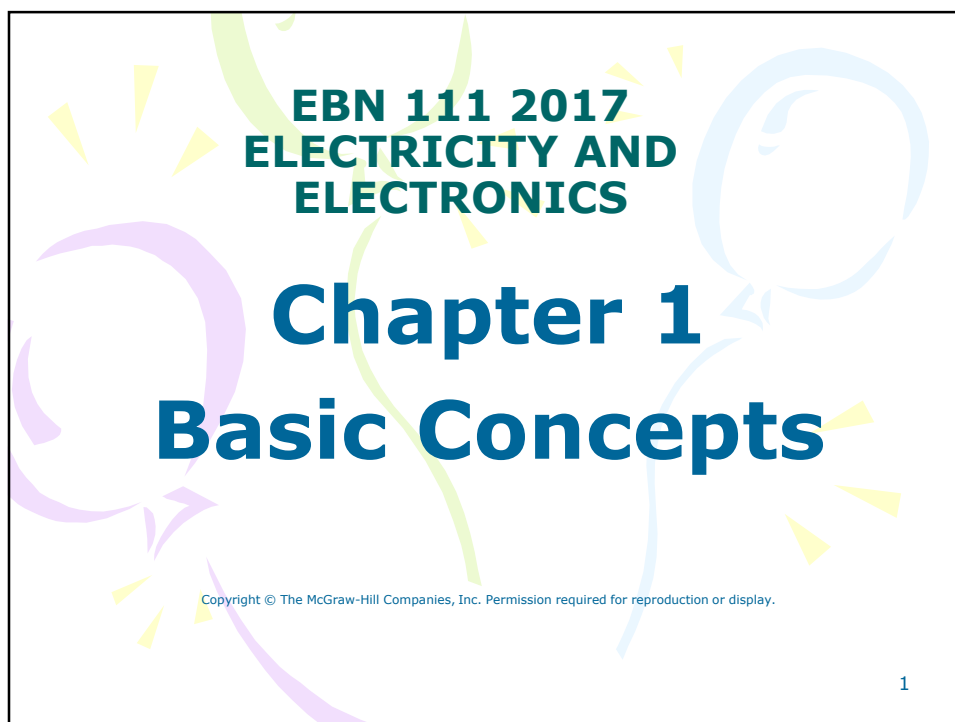




Chapter 1: Lecture #1

1.1 System of Units (1)

Six basic units: SI Units

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd

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1.1 System of Units (2)

The derived units commonly used in electric circuit theory

Decimal multiples and submultiples of SI units

Quantity	Unit	Symbol
electric charge	coulomb	C
electric potential	volt	V
resistance	ohm	Ω
conductance	siemens	S
inductance	henry	H
capacitance	farad	F
frequency	hertz	Hz
force	newton	N
energy, work	joule	J
power	watt	W
magnetic flux	weber	Wb
magnetic flux density	tesla	T

Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

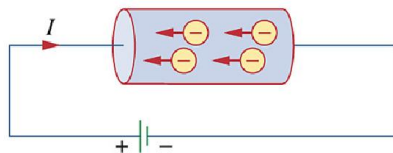
1.2 Electric Charge

- **Charge** is an electrical property of the atomic particles of which matter consists, measured in **coulombs (C)**.
- The charge **e** on one electron is negative and equal in magnitude to $1.602 \times 10^{-19} \text{ C}$ which is called an electronic charge.
- $1 \text{ C} = 1/e = 6.24 \times 10^{18}$ electrons
- **Law of conservation of Charge:** Charge can neither be created nor destroyed, it can only be transferred.

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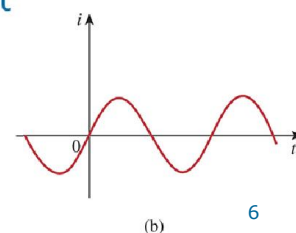
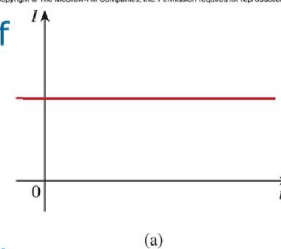
1.3 Current (2)

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- Electric current $i = dq/dt$. The unit of ampere can be derived as $1 \text{ A} = 1 \text{ C/s}$.
- Amount of charge transferred between t_0 and t : $Q \triangleq \int_{t_0}^t i \, dt$
- A **direct current (dc)** is a current that remains constant with time.
- An **alternating current (ac)** is a current that varies sinusoidally with time. (reverse direction)

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Current

- If DC
 - $Q = I\Delta t$
- If AC
 - Integral

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1.4 Voltage

- Voltage (or potential difference) is the **energy (work)** required to move a **unit charge** through an element, measured in volts (V).
- Work is done by the electromotive force (**emf**) aka **voltage** or **potential difference**.
- Mathematically, (volt)
 - w is energy in joules (J) and q is charge in coulomb (C).
 - voltage is change in energy per charge

$$v_{ab} = dw / dq$$

$$v_{ab} = v_a - v_b$$

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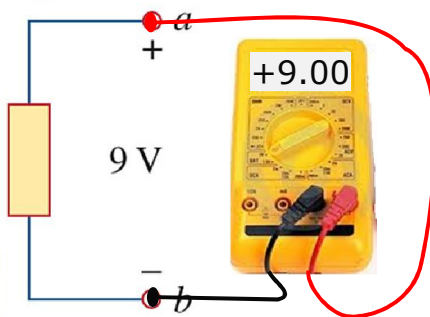
1.4 Voltage

Electric voltage, v_{ab} , is always across the circuit element or between two points in a circuit.

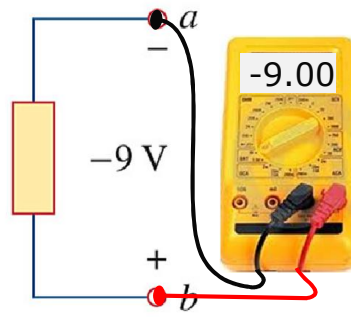
$v_{ab} > 0$ means the potential of **a** is higher than potential of **b**.

$v_{ab} < 0$ means the potential of **a** is lower than potential of **b**.

$$v_{ab} = -v_{ba}$$



(a) $v_{ab} = +9V$



(b) $v_{ba} = -9V$

Voltage

- Assume

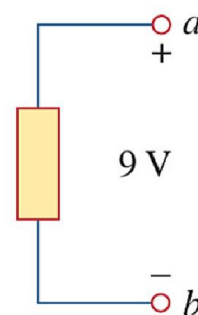
- $-V_a = 9V$

- $-V_b = 0V$

- $V_{ab} = 9 - 0 = 9V$

- $V_{ba} = 0 - 9 = -9V$

- $V_{ab} = -V_{ba}$



(a)